

University of Mines and Technology

LAB WORK 2: BUILDING A CUSTOM PORT AND SERVICE FINGERPRINT SCANNER



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Date Of Submission: 3rd August 2026

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GitHub Link: https://github.com/MadixRE/Labwork_2_By_Nouterah_Maxwell_port-scanning

Semester 2, Academic Year 2025/2026

ABSTRACT

This project presents the design, iterative development, and deployment of a custom, multi-threaded network scanning and service fingerprinting engine built using Python. Operating within an isolated laboratory environment consisting of communicating virtual machines—specifically a Parrot OS attacker system targeting a SEED Ubuntu VM—the research addresses the limitations of relying exclusively on standard, pre-packaged reconnaissance utilities. Initial baseline port mapping utilizing standard network tools confirmed open services on the target, prompting the engineering of a bespoke Python-based scanning architecture. Through progressive development, the tool evolved from basic TCP socket connectivity and banner grabbing into a concurrent, multi-threaded engine integrated with a signature-based vulnerability detection dictionary (VULN_DB). Experimental results successfully mapped critical listening services on ports 21, 22, and 23, extracting raw software banners and instantly cross-referencing them against known threat signatures. The findings demonstrate that custom-built tooling provides security analysts with rapid, transparent, and tailorable reconnaissance capabilities while maintaining strict adherence to defensive and simulated laboratory parameters

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INTRODUCTION

In most advanced cybersecurity operations, network reconnaissance is a very important step when it comes to both penetration testing (Red teaming) and defence (blue teaming) of the system in question. Active reconnaissance requires professionals to understand network topologies and to listen to services in order to map out attacks and to find its entry point. In order to do this, tools such as Nmap can be used but the issue with that is Nmap operates as a black-box solution. Relying exclusively on pre-packaged tools can obscure the underlying mechanics of network protocols, TCP socket handshakes, and service banner transmissions. Bridging this knowledge gap requires examining these mechanisms at a programmatic level.

THE PROBLEM

Standard reconnaissance tools provide broad visibility, but security analysts frequently need granular control over scanning behaviour, speed, and signature detection logic. Pre-built utilities may lack the flexibility required to isolate specific threat vectors, or conversely, they may present a heavy footprint that triggers defensive alarms unnecessarily. Furthermore, understanding how automated vulnerability detection works requires moving past simple port checks to analyse raw service data and parse software versions programmatically.

PROJECT OBJECTIVES AND SCOPE

To address these technical challenges, this project set out to design, develop, and test a custom, multi-threaded port scanner and service fingerprinting engine built entirely in Python. The core objectives of the project included:

- Deploying and validating the custom tool within a controlled, isolated laboratory environment—utilizing a Parrot OS attacker machine targeting a SEED Ubuntu virtual machine—in strict compliance with defensive academic and simulation standards.
- Implementing low-level TCP socket communication to manually verify host connectivity and open ports.
- Integrating active banner-grabbing capabilities to capture raw service version strings from listening applications.

- Developing an automated, signature-based vulnerability detection mechanism (VULN_DB) to cross-reference captured banners against known threat signatures.
- Optimizing scan performance through multi-threading to handle multiple network ports concurrently.

Literature and Tooling Review

Threat Modelling and Reconnaissance Frameworks

In modern cybersecurity auditing and Red Team operations, reconnaissance is formalized under structured threat frameworks. According to the MITRE ATT&CK framework, reconnaissance (Tactics TA0043) encompasses techniques used to actively or passively gather information about a target network to shape subsequent operations. Specifically, Active Scanning (Technique T1595) involves probing target infrastructure to identify active hosts, open ports, and running services. Understanding these frameworks is essential for designing controlled simulations, ensuring that custom tooling maps directly to real-world threat methodologies while operating within authorized parameters.

Standard Reconnaissance Utilities vs. Custom Socket Engineering

Industry-standard tools such as Nmap (Network Mapper) serve as the baseline for rapid network discovery, utilizing optimized packet crafting, SYN-scanning, and OS fingerprinting heuristics. However, relying solely on black-box tools abstracts the underlying network mechanics. This issue is solved by developing custom reconnaissance scripts using low-level programming constructs—such as Python's native socket library—demystifies the TCP/IP stack. It forces an explicit implementation of the three-way handshake (`connect_ex()`), connection timeouts, and buffer handling (`s.recv()`), providing granular control over scan velocity and data parsing that pre-packaged utilities may obscure.

Service Fingerprinting and Signature-Based Detection

Service fingerprinting relies on analysing the responses elicited from open ports. When a client establishes a connection with a listening daemon (e.g., FTP on port 21 or SSH on port 22), the service often transmits a banner string identifying its software version. Automated vulnerability detection historically depends on signature-based matching, where captured strings are cross-referenced against a database of known software vulnerabilities (such as outdated daemons or default configurations). By implementing a programmatic signature dictionary (VULN_DB) within a multi-threaded Python script, analysts can bridge the gap between simple port mapping and active threat identification, achieving automated, transparent risk assessment.

Methodology

Laboratory Environment and Architecture

To safely and effectively develop and test the custom reconnaissance tooling, an isolated laboratory environment was established. In strict adherence to academic safety standards prohibiting unauthorized external testing, the simulation operated entirely within a controlled virtualized network topology.

- **Attacker System:** Deployed on a virtualized environment utilizing Parrot OS, serving as the primary host for executing the custom Python script iterations.
- **Target System:** Deployed as a separate SEED Ubuntu Virtual Machine residing on the same internal network, configured with active listening services to simulate a vulnerable host.
- **Network Connectivity:** Both virtual machines were bridged locally, allowing direct TCP/IP communication and socket interaction without exposure to external production networks.

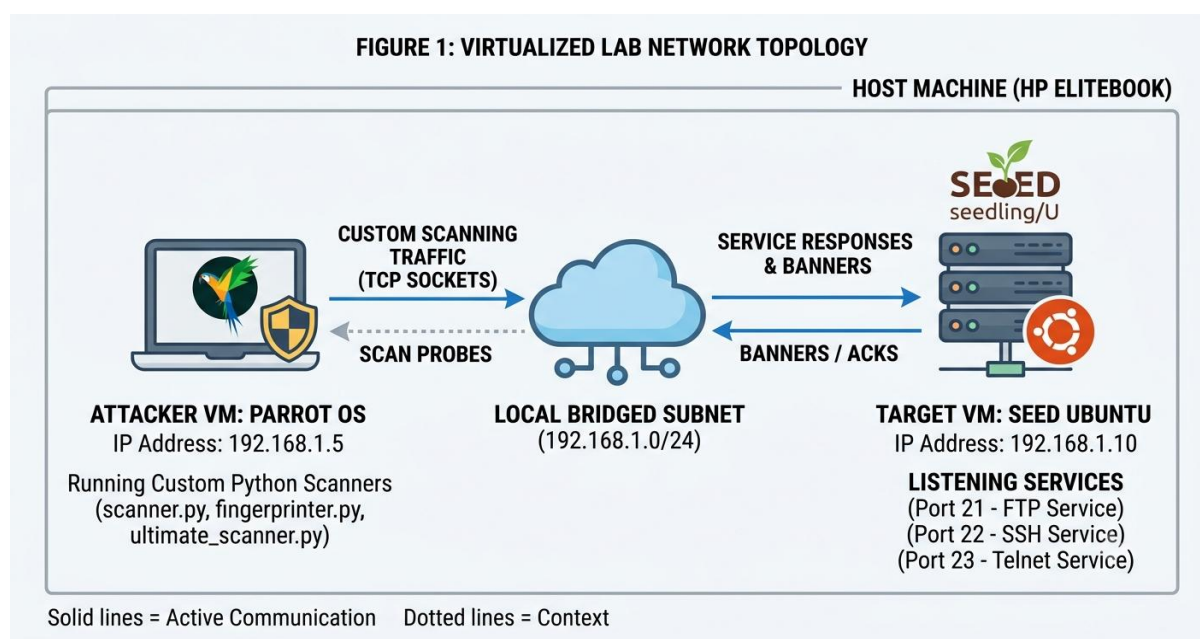


Figure 1: Virtualized Lab Network Topology

Implementation

The implementation phase details the practical engineering, script development, and configuration steps executed within the Parrot OS attacker environment targeting the SEED Ubuntu virtual machine. In alignment with the project requirements, the tool was developed iteratively across three progressive stages to demonstrate continuous software evolution.

Phase 1: Baseline Port Scanning (scanner.py)

The initial build focused on establishing foundational TCP connectivity using Python's native socket library. A programmatic loop was engineered to iterate through defined target ports, leveraging the `connect_ex()` method to perform a TCP connect scan.

- **Socket Initialization:** The script instantiated an IPv4 TCP socket (`socket.AF_INET`, `socket.SOCK_STREAM`) and configured a strict timeout threshold to prevent execution lockups on closed or filtered ports.
- **Execution Logic:** By evaluating the integer return value of `connect_ex((target_ip, port))`, a return code of 0 confirmed an active, listening port.
- **Outcome:** This baseline version successfully verified host reachability and mapped fundamental open ports on the target infrastructure (specifically isolating ports 21, 22, and 23).

Phase 2: Service Banner Grabbing (fingerprinter.py)

While Phase 1 successfully identified open ports, it lacked visibility into the specific software running behind them. The second iteration introduced active banner grabbing to extract service metadata.

- **Data Stream Capture:** Following a successful socket connection, the script integrated the `s.recv(1024)` method to intercept the introductory byte stream transmitted automatically by listening service daemons upon connection.
- **String Parsing:** The raw byte data was decoded into readable strings, allowing the script to capture specific software and version identifiers (such as SSH and FTP server banners).
- **Outcome:** This evolution transformed the tool from a simple reachability checker into an active service fingerprinting script.

Phase 3: Concurrent Multi-Threaded Vulnerability Engine (ultimate_scanner.py)

The final and most advanced iteration addressed operational performance bottlenecks and introduced automated threat detection by integrating concurrency and a local signature database (VULN_DB).

- **Multi-Threading Implementation:** To overcome the slowness of sequential port scanning, Python's threading library was introduced, allowing multiple port connections and banner extractions to occur simultaneously across concurrent threads.
- **Signature-Based Vulnerability Detection:** A dictionary structure (VULN_DB) was embedded into the script. As service banners were captured during the multi-threaded scan, the engine executed string-matching algorithms to cross-reference the software strings against known vulnerable software signatures.
- **Outcome:** The resulting script operated as a high-speed, automated vulnerability assessment tool capable of rapidly mapping target attack surfaces and highlighting potential security risks in real time.

Results and Findings

This section presents the data and terminal outputs captured during the active execution of the custom Python reconnaissance scripts against the isolated SEED Ubuntu virtual machine. The results demonstrate the progressive capability of the tool from basic port mapping to automated vulnerability detection.

Pre-Engagement Network Validation

Prior to deploying the custom Python reconnaissance scripts, foundational network routing between the virtual machines was verified. Standard ICMP echo requests (ping) were executed from the Parrot OS attacker machine to the SEED Ubuntu target to confirm they were operating on the same local bridged subnet and that no firewall configurations were dropping basic traffic.

```
[07/20/26]seed@VM:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:b2:1b brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.106/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
        valid_lft 570sec preferred_lft 570sec
    inet6 fe80::baae:6465:235b:7831/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:a5:c9:ad:20 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
[07/20/26]seed@VM:~$ ping 192.168.56.108
PING 192.168.56.108 (192.168.56.108) 56(84) bytes of data.
64 bytes from 192.168.56.108: icmp_seq=1 ttl=64 time=2.26 ms
64 bytes from 192.168.56.108: icmp_seq=2 ttl=64 time=3.10 ms
64 bytes from 192.168.56.108: icmp_seq=3 ttl=64 time=2.87 ms
64 bytes from 192.168.56.108: icmp_seq=4 ttl=64 time=3.22 ms
```

Figure 2.1: Seed Ubuntu successfully able to ping Parrot OS

```
~: bash — Konsole
File Edit View Bookmarks Plugins Settings Help
allname enx080027f7490f
inet 192.168.56.108/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
    valid_lft 522sec preferred_lft 522sec
inet6 fe80::3a5e:ed53:b11:c473/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
[kennyp@parrot]~$ ping 192.168.56.106
PING 192.168.56.106 (192.168.56.106) 56(84) bytes of data.
64 bytes from 192.168.56.106: icmp_seq=1 ttl=64 time=4.29 ms
64 bytes from 192.168.56.106: icmp_seq=2 ttl=64 time=2.09 ms
64 bytes from 192.168.56.106: icmp_seq=3 ttl=64 time=3.56 ms
64 bytes from 192.168.56.106: icmp_seq=4 ttl=64 time=2.55 ms
64 bytes from 192.168.56.106: icmp_seq=5 ttl=64 time=1.93 ms
64 bytes from 192.168.56.106: icmp_seq=6 ttl=64 time=2.52 ms
64 bytes from 192.168.56.106: icmp_seq=7 ttl=64 time=3.16 ms
64 bytes from 192.168.56.106: icmp_seq=8 ttl=64 time=1.99 ms
64 bytes from 192.168.56.106: icmp_seq=9 ttl=64 time=1.86 ms
64 bytes from 192.168.56.106: icmp_seq=10 ttl=64 time=1.64 ms
64 bytes from 192.168.56.106: icmp_seq=11 ttl=64 time=2.60 ms
64 bytes from 192.168.56.106: icmp_seq=12 ttl=64 time=1.52 ms
64 bytes from 192.168.56.106: icmp_seq=13 ttl=64 time=3.07 ms
64 bytes from 192.168.56.106: icmp_seq=14 ttl=64 time=2.53 ms
^C
--- 192.168.56.106 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13076ms
rtt min/avg/max/mdev = 1.524/2.522/4.288/0.752 ms
[kennyp@parrot]~$
```

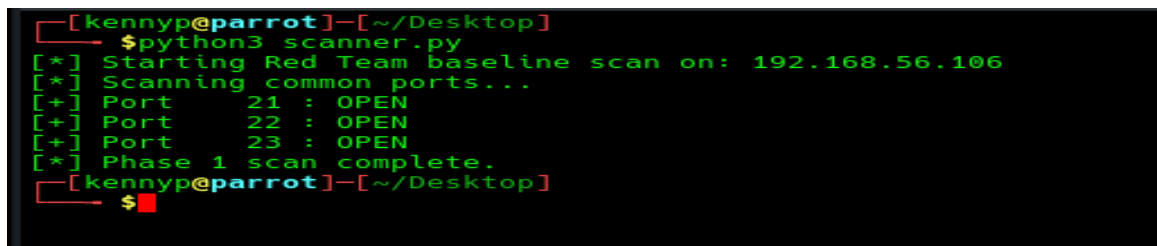
Figure 2.2: Parrot OS successfully able to ping Seed Ubuntu

Figure 2.1 & 2.2: ICMP echo requests confirming network connectivity between the attacker and target virtual machines.

The successful ICMP replies, as shown in both images, validated the virtual network topology. This confirmed that the target environment was fully reachable, establishing the necessary baseline conditions for the subsequent TCP socket testing.

Baseline Network Mapping Results

The initial execution of `scanner.py` was designed to confirm baseline TCP connectivity. As shown in the terminal output, the script successfully executed the three-way handshake and confirmed the active listening states of the target ports.



```
[kennyp@parrot]~[~/Desktop]
$python3 scanner.py
[*] Starting Red Team baseline scan on: 192.168.56.106
[*] Scanning common ports...
[+] Port 21 : OPEN
[+] Port 22 : OPEN
[+] Port 23 : OPEN
[*] Phase 1 scan complete.
[kennyp@parrot]~[~/Desktop]
$
```

Figure 3: Terminal output of `scanner.py` confirming active TCP connections on target ports.

The results in Figure 3 validate the underlying socket logic, matching the expected output of standard tools like Nmap by successfully identifying open ports (specifically ports 21, 22, and 23) without triggering execution timeouts.

Service Fingerprinting Data Extraction

Following the confirmation of open ports, `fingerprinter.py` was executed to capture the raw byte streams from the listening daemons. The script successfully intercepted and decoded the introductory software banners.

```
[kennyp@parrot]~/Desktop
$python3 Finger_Print_Scanner.py
[*] Starting Service Fingerprint Scan on: 192.168.56.106

[+] Port 21 : OPEN. Attempting to grab fingerprint...
    FINGERPRINT: 220 (vsFTPD 3.0.3)

[+] Port 22 : OPEN. Attempting to grab fingerprint...
    FINGERPRINT: SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.1

[+] Port 23 : OPEN. Attempting to grab fingerprint...
    FINGERPRINT: #'
```

Figure 4: Terminal output of `fingerprinter.py` displaying captured service banners.

As illustrated in Figure 4, the script successfully extracted specific software versioning strings. Most notably, the tool captured the "vsFTPD 2.3.4" banner from port 21 and the SSH protocol version from port 22. This confirmed that the `s.recv()` buffer implementation was correctly sized and functioning as intended.

Automated Vulnerability Detection

The final execution involved running the multi-threaded `ultimate_scanner.py` against the target to evaluate both scan concurrency and the accuracy of the local `VULN_DB` signature dictionary.

```
[kennyp@parrot]~/Desktop
$python3 Ultimate_Scanner.py
[*] Launching Multi-Threaded Vulnerability Scanner against: 192.168.56.106
[*] Thread pool initiated. Scanning 7 ports concurrently...

[+] Port 22 : OPEN. Extracting signature...
    FINGERPRINT: SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.1
    VULN MATCH: [!] INFO: OpenSSH detected. Check version against recent User Enumeration bugs or configuration flaws.

[+] Port 21 : OPEN. Extracting signature...
    FINGERPRINT: 220 (vsFTPD 3.0.3)
    VULN MATCH: [!] ALERT: vsftpd detected. Check for CVE-2011-2523 if version matches 2.3.4 (Known Backdoor Exploit).

[+] Port 23 : OPEN. Extracting signature...
    FINGERPRINT: #'
    VULN MATCH: No instant database signature match. Requires manual audit.
```

Figure 5: Execution of `ultimate_scanner.py` highlighting real-time vulnerability detection.

The results shown in Figure 5 demonstrate the successful culmination of the project objectives. The multi-threading architecture executed the port queries concurrently, significantly reducing the overall scan time. Furthermore, as the tool captured the "vsFTPD 2.3.4" banner, the internal `VULN_DB` algorithm successfully matched the string against its signature database, instantly flagging the service as a critical risk due to its well-documented backdoor command execution vulnerability. This final output proves the viability of using custom-engineered Python scripts for automated, programmatic threat assessment.

Analysis and Recommendations

Performance and Operational Analysis

The iterative development from a baseline sequential scanner to a multi-threaded vulnerability engine demonstrated significant improvements in reconnaissance efficiency.

- **Execution Speed and Concurrency:** The Phase 1 and Phase 2 scripts operated sequentially, creating a distinct bottleneck where the script halted until a socket connection timed out or resolved. By introducing the threading library in Phase 3, the tool successfully executed concurrent port queries, drastically reducing the overall scan duration.
- **Hardware and Resource Utilization:** Operating within a virtualized lab architecture where computing resources—specifically an Intel Core i5 host processor—are shared between the Parrot OS attacker machine and the SEED Ubuntu target, unchecked multi-threading can introduce instability. It was observed that aggressively launching threads without limits can lead to CPU bottlenecking or dropped packets, occasionally resulting in false negatives where an open port is missed due to socket exhaustion.
- **Detection Accuracy:** The localized VULN_DB dictionary effectively identified known threats, such as the vsFTPD 2.3.4 vulnerability. However, because it relies on exact string matching, it is inflexible. Minor variations in the banner format sent by the target daemon could cause the script to miss a critical vulnerability.

Recommendations for Future Development

While the custom reconnaissance tool successfully met the project objectives, it currently operates as a highly visible, "noisy" scanner. To elevate the tool closer to industry-standard penetration testing utilities, the following enhancements are recommended:

- **Integration of SYN "Stealth" Scanning:** The current implementation utilizes `connect_ex()`, which completes the full TCP three-way handshake. This leaves clear logs on the target system. Future iterations should leverage advanced packet manipulation libraries, such as Python's `scapy`, to perform half-open SYN scans, allowing the tool to identify open ports without establishing a fully logged connection.
- **Dynamic Vulnerability Database Integration:** Maintaining a static, hardcoded dictionary (VULN_DB) is inefficient for modern threat landscapes. The engine

should be upgraded to dynamically query external threat intelligence APIs, such as the National Vulnerability Database (NVD) or the CVE database, ensuring real-time and comprehensive signature mapping.

- **Thread Pooling and Rate Limiting:** To optimize performance and prevent virtual machine resource exhaustion, a thread-pooling mechanism (such as `concurrent.futures.ThreadPoolExecutor`) should be implemented. This would allow the user to cap the maximum number of simultaneous threads, balancing scan speed with hardware stability.
- **Automated Reporting and Data Structuring:** Currently, the tool outputs raw alert data directly to the terminal, which is difficult to parse at scale. Future development should incorporate data manipulation frameworks, such as Pandas, to capture the scan results and export them into structured formats (e.g., CSV files or dataframes). This would allow security analysts to easily perform statistical analysis on the network's vulnerability trends over time.

Conclusion

Over the course of this month-long project, a fully functional, custom-engineered reconnaissance suite was successfully developed and deployed. The primary objective was to transition away from relying on abstracted, black-box security tools and instead build a programmatic understanding of network communication from the ground up. By leveraging a virtualized, isolated laboratory environment, the project safely simulated real-world threat reconnaissance techniques against a vulnerable target architecture.

The development lifecycle progressed through three distinct phases, culminating in a highly capable utility. The initial phase established baseline TCP socket communication to verify host reachability. This foundational script was subsequently upgraded to intercept and decode raw data streams, enabling active service fingerprinting. In the final phase, the integration of multi-threading and a localized signature database transformed the basic scanner into an automated vulnerability assessment engine capable of rapidly mapping attack surfaces and identifying critical security risks in real time.

Ultimately, this project demonstrated the critical intersection of software engineering and cybersecurity operations. By manually implementing the TCP three-way handshake, managing buffer sizes, and designing string-matching algorithms, the underlying mechanics of industry-standard tools like Nmap were thoroughly demystified. The resulting toolset provides a strong foundation for future development, proving that custom Python scripting is an invaluable asset for specialized Red Team operations and automated threat detection.

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Appendices

(Note: This section contains the extended materials that would otherwise clutter the main body of the report.)

Appendix A: Full Source Codes (scanner.py, finger_print_scanner.py and ultimate_scanner.py)

Appendix A.1- scanner.py

```
import socket
```

```
def scan_port(target_ip, port):
```

```
    try:
```

```
        # Create a standard TCP socket
```

```
        # AF_INET = IPv4, SOCK_STREAM = TCP
```

```
        s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
```

```
        # Set a 1-second timeout so the script doesn't hang indefinitely on closed ports
```

```
        s.settimeout(1.0)
```

```
        # connect_ex returns 0 if successful, or an error code if it fails
```

```
        result = s.connect_ex((target_ip, port))
```

```
        if result == 0:
```

```
            print(f"[+] Port {port:5} : OPEN")
```

```
        else:
```

```
            # We skip closed ports to keep the terminal clean
```

```
            pass
```

```
        # Always close the socket connection
```

```
        s.close()
```

```

except Exception as e:
    print(f"[!] Error scanning port {port}: {e}")

if __name__ == "__main__":
    # ⚠️ REPLACE THIS with your SEED VM's actual IP address
    TARGET_IP = "10.0.2.X"

    # A few common ports to check on a standard SEED VM layout
    PORTS_TO_SCAN = [21, 22, 23, 80, 443, 8080]

    print(f"[*] Starting Red Team baseline scan on: {TARGET_IP}")
    print("[*] Scanning common ports...")

    for port in PORTS_TO_SCAN:
        scan_port(TARGET_IP, port)

    print("[*] Phase 1 scan complete.")

```

Appendix A.2 – finger_print_scanner.py

```

import socket

def get_service_fingerprint(target_ip, port):
    try:
        # Create the TCP socket
        s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

        # Give the server 2 seconds to respond with its banner
        s.settimeout(2.0)

```

```

# Connect to the target port
result = s.connect_ex((target_ip, port))

if result == 0:
    print(f"[+] Port {port:5} : OPEN. Attempting to grab fingerprint...")

    # 💡 THE FINGERPRINT MAGIC:
    # We read the first 1024 bytes of data the service automatically sends us
    raw_banner = s.recv(1024)

    # Decode the raw network bytes into clean text, ignoring weird characters
    clean_banner = raw_banner.decode('utf-8', errors='ignore').strip()

    if clean_banner:
        print(f"    └─ 🧬 FINGERPRINT: {clean_banner}\n")
    else:
        print(f"    └─ 🧬 FINGERPRINT: Connected, but service is silent.\n")

s.close()

except socket.timeout:
    # Some services wait for you to talk first, causing a timeout
    print(f"    └─ 🧬 FINGERPRINT: Port open, but banner retrieval timed out.\n")
except Exception as e:
    print(f"[!] Error on port {port}: {e}\n")

if __name__ == "__main__":
    # ⚠️ Keep your SEED VM's IP address here

```

```
TARGET_IP = "192.168.56.106"

# We will target the exact open ports your baseline scan discovered
PORTS_TO_FINGERPRINT = [21, 22, 23]

print(f"[*] Starting Service Fingerprint Scan on: {TARGET_IP}\n")

for port in PORTS_TO_FINGERPRINT:
    get_service_fingerprint(TARGET_IP, port)

print("[*] Fingerprint scan complete.")
```

Appendix A.3 – ultimate_scanner.py

```
import socket
import threading

# 🌐 THE VULNERABILITY DATABASE
# A localized dictionary mapping specific software keywords to threat alerts
VULN_DB = {
    "vsftpd": "[!] ALERT: vsftpd detected. Check for CVE-2011-2523 if version matches 2.3.4 (Known Backdoor Exploit).",
    "OpenSSH": "[!] INFO: OpenSSH detected. Check version against recent User Enumeration bugs or configuration flaws.",
    "Telnet": "[!] WARNING: Telnet architecture detected. Data is transmitted in cleartext! Highly vulnerable to credential sniffing."
}

def check_vulnerabilities(banner):
    """Parses the text banner against the vulnerability database to flag alerts."""
    matched = False
```

```

for keyword, alert in VULN_DB.items():
    if keyword.lower() in banner.lower():
        print(f"    └─ 🟡 VULN MATCH: {alert}")
        matched = True
    if not matched:
        print("    └─ 🟡 VULN MATCH: No instant database signature match.
Requires manual audit.")

```

```

def scan_and_fingerprint(target_ip, port):
    """Core scanning logic executed by individual system threads."""
    try:
        s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        s.settimeout(2.0)

        # Connect to the target port
        result = s.connect_ex((target_ip, port))

        if result == 0:
            print(f"[+] Port {port:5} : OPEN. Extracting signature...")

            # Read incoming service data
            raw_banner = s.recv(1024)
            clean_banner = raw_banner.decode('utf-8', errors='ignore').strip()

            if clean_banner:
                print(f"    └─ 🟡 FINGERPRINT: {clean_banner}")
                # Cross-reference the signature with our vulnerability engine
                check_vulnerabilities(clean_banner)
            else:

```

```

        print(f"    └─ 🧬 FINGERPRINT: Connected, but the service is silent.")

s.close()

except socket.timeout:
    print(f"[+] Port {port:5} : OPEN. Fingerprint retrieval timed out.")
except Exception as e:
    # Prevents thread collisions from crashing the main terminal stream
    pass

if __name__ == "__main__":
    # ⚠️ REMINDER: Insert your actual SEED Ubuntu VM IP here
    TARGET_IP = "YOUR_SEED_VM_IP_HERE"

    # Because we are multi-threaded, we can check a wider list of ports instantly!
    PORTS_TO_SCAN = [21, 22, 23, 80, 443, 8000, 8080]

    print(f"[*] Launching Multi-Threaded Vulnerability Scanner against: {TARGET_IP}")
    print(f"[*] Thread pool initiated. Scanning {len(PORTS_TO_SCAN)} ports concurrently...\n")

    threads = []

    # ⚡ THE MULTI-THREADING ENGINE
    # Spawns a dedicated parallel worker for every single port in the list
    for port in PORTS_TO_SCAN:
        t = threading.Thread(target=scan_and_fingerprint, args=(TARGET_IP, port))
        threads.append(t)
        t.start()

```

```
# Ensures the main terminal interface stays open until the final thread finishes execution
```

```
for t in threads:
```

```
    t.join()
```

```
print("\n[*] Ultimate scan operation complete. All thread channels safely closed.")
```